Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	06 July 2026
Team ID	
Project Name	Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Data Ingestion & Cleaning	Import UNESCO heritage site CSV data; remove duplicates and null/invalid rows; standardize country, region and date fields
FR-2	Site Distribution Visualization	Bar chart of sites by category (Cultural/Mixed/Natural); Pie chart of site count per region; Treemap of sites by country
FR-3	Geographic Visualization	World map showing top 10 regions/sites by area; Tooltips showing country, region and site details on hover
FR-4	Trend Analysis	Line chart of yearly inscriptions with Actual vs Estimate forecast; Line chart of regional inscription trends split by region
FR-5	Risk & Danger Analysis	Pie chart of sites In Danger vs Not in Danger; List/bar of Top 10 danger-prone sites; Map of heritage-ended sites
FR-6	Dashboard Assembly	Combine all charts into an Overview dashboard and a Danger Analysis dashboard with consistent titles, borders and layout
FR-7	Story Narrative	Build a Tableau Story that walks a viewer through the Overview, Trends and Danger Analysis dashboards in sequence
FR-8	Web Publishing	Publish dashboards and story to Tableau Public; generate embeddable share/embed codes for each
FR-9	Web Application Integration	Embed published Tableau dashboards/story into a Flask web application with navigation, dark/light mode toggle

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Dashboards use clear titles, consistent color coding, tooltips, and a simple navigation bar so users can explore heritage data without training
NFR-2	**Security**	Published Tableau content is read-only for public viewers; no personal or sensitive data is exposed since the dataset is public UNESCO information
NFR-3	**Reliability**	Dashboards are backed by a cleaned, validated dataset with no missing critical fields, so charts render consistently every time they are loaded
NFR-4	**Performance**	Dashboards and the embedded Flask page load within a few seconds on a standard broadband connection; charts render without noticeable lag when filtering
NFR-5	**Availability**	Dashboards are hosted on Tableau Public (99%+ uptime) and the Flask app can be run locally or deployed, ensuring the project is accessible for demonstration at any time
NFR-6	**Scalability**	The workbook structure and data model can accommodate additional years, regions, or new UNESCO sites being added without redesigning the dashboards
NFR-7	**Maintainability**	Worksheets are clearly named (e.g., Heritage Sites by Country, Regional Inscription Trends) so future updates or new scenario charts can be added easily
NFR-8	**Portability**	The Flask web app and embed codes work across modern browsers (Chrome, Edge, Firefox) and are responsive to different screen sizes